Umair Javaid Manj

Engineer working on spinal pain, neurophysiology and machine learning

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Research portfolio and video | 0009-0008-7085-3662 | /umairjm

Research Experience

Doctoral Researcher, Chronic Low Back Pain Oct 2025 to Present | Auckland, NZ
Auckland University of Technology, Health & Artificial Intelligence

- Building an **AI based biomarker** for identifying chronic lower back pain.
- Designed a **segmental vibrotactile EEG protocol**: tactor pairs on the painful and a healthy spinal segment of the same person, cortical response compared between the two.
- Ran **Pilot 1** (one participant, 75 placements). Traced a first pass on versus off decoding result (92 to 99%) to **unequal analysis windows** and withdrew it; duration matched, the effect was zero.
- Set the fixes for the next recording: measure the vibration drive frequency before the session and write the stimulus trigger into the EEG file.
- Produced a two minute **protocol animation** (participant setup, tactor placement, expected segmental EEG contrast). watch

Data & AI Governance Analyst Aug 2026 to Present | Auckland, NZ
Centre for Chiropractic Research, New Zealand College of Chiropractic

- Collect and analyse data on the Centre's **chiropractic research studies**.
- Responsible for how **participant data** is classified, stored, shared and disposed of under the Privacy Act 2020.

Research Associate / Research Assistant Aug 2021 to Apr 2023 | Islamabad, PK
COGNET Lab, SEECS, NUST

- Worked on defending **deep learning models** against adversarial attacks and on post quantum cryptography.
- Built an end to end auditable system combining cryptography with **computer vision**.
- Mentored the lab's junior research assistants.

Instructor and Program Manager May 2023 to Oct 2025 | Islamabad, PK
NUST

- Supervised **18 student term projects**; named Best Instructor.
- Ran a national training program in AI, networking and cybersecurity, from scheduling through delivery to reporting.

Earlier: Network Engineer, Nexalink, Islamabad (2016 to 2017).

Publications

Published

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). *Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review*. Medical Engineering & Physics. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). *Sleep disorder prediction with PSO-optimized machine and deep learning models*. ICODT2, pp. 1 to 6. doi
- Haq, M. I. U., Ahmad, Z., **Manj, U. J.**, Akhter, M. N., Arfeen, Z. A., & Qamar, S. D. (2026). *Bio-inspired soft robotic gripper with variable stiffness using magneto-rheological control*. ICRAI, pp. 1 to 6. doi
- Haq, M. I. U., **Manj, U. J.**, Paracha, S. Q., Imam, Y., Rizvi, D. Q., & Farooqi, A. R. (2026). *Neuro-symbolic AI-driven secure communication framework for autonomous cyber-physical systems*. APAN 2026, pp. 1 to 6. doi

Under review, biomedical

- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. *Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU*. IEEE Transactions on Biomedical Engineering.
- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. *Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test*. Medical & Biological Engineering & Computing.
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. *Gated feature fusion and age confounding in plantar thermography for diabetes screening*. Biomedical Physics & Engineering Express.
- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. *Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease*. Machine Learning: Health.

Under review, other: five further manuscripts on evaluation of intrusion detection (2), FPGA spectrum inference, thermal fire detection and a driver behaviour dataset.

Education

PhD, Health & Artificial Intelligence — Oct 2025 to Present | Auckland, NZ
Auckland University of Technology

- AI based biomarker for chronic lower back pain; segmental EEG protocol and pilot (see above).

MS, Computational Science & Engineering — 2017 to 2021 | Islamabad, PK
National University of Sciences and Technology (NUST)

- Thesis: energy optimisation in data centres using **fuzzy inference systems** (~30% cost reduction).

BE, Computer Engineering — 2012 to 2016 | Islamabad, PK
National University of Sciences and Technology (NUST)

- Thesis: NATO narrowband waveform on a software defined radio kit (convolutional encoding, phase modulation).

Skills

- **Signals & neurophysiology:** EEG and IMU signal analysis, vibrotactile stimulation protocols, spectral and time frequency analysis.
- **Machine learning & statistics:** Python, PyTorch, scikit-learn; subject level evaluation, leakage and confounding checks.
- **Programming:** Python, C++, Bash, PowerShell.
- **Research practice:** participant data governance (Privacy Act 2020, ISO/IEC 27001), systematic review, scientific writing.

Referees

- **Dr. Nazia Perwaiz**, MSCA Industrial Postdoctoral Fellow, School of Computer Science, University of Galway, Ireland; Assistant Professor, SEECS, NUST. nazia.perwaiz@seecs.edu.pk
- **Dr. Haider Ali**, Faculty of Design and Creative Technologies, Auckland University of Technology; AI Engineer, KPMG New Zealand. haider.ali@autuni.ac.nz